

IDS2Eval Scorecard

bot-iot-official

16 ok · 3 warning · 3 flag · 22 checks

REVIEW REQUIRED

- **dedup_check**
train duplicates: 0 (0.00%); test rows leaking a train feature-match: 0 (0.00%); test-internal duplicates: 0

ok
- **label_conflict_check**
no feature vector maps to more than one label

ok
- **near_duplicate_class_check**
no near-zero-distance feature vectors found across 5 classes tested

ok
- **leakage_screen**
no single feature or pair dominates importance (top1=29.4%, top2=48.4%)

ok
- **one_rule_check**
single rule 'ltime <= 1.528e+09' reaches 97.5% test accuracy (97.5% train). This problem may be solvable without learning attack behavior

flag
- **feature_auc_ranking_check**
'ltime' alone reaches AUC 1.000 identifying 'Theft'; a single feature this separable is worth checking for a leakage artifact

flag
- **identity_column_flag**
standalone AUC by column: {'saddr': 0.522, 'daddr': 0.865, 'sport': 0.409, 'dport': 0.536}. Suggest dropping: ['daddr']

flag
- **port_protocol_shortcut_check**
standalone AUC of 'sport' + 'proto' combined: 0.481

ok
- **temporal_leakage_check**
standalone AUC of 'stime' alone: 0.500

ok
- **temporal_realism_check**
timestamp column has no usable time span to test

ok
- **flow_group_leakage_check**
no schema.flow_id_columns configured

ok
- **row_order_leakage_check**
adjacent-row label-transition rate (train=1.00, test=1.00) as a share of what a randomly shuffled ordering would produce

ok
- **homogeneity_test**
4 classes tested; test-to-train and train-internal proximity statistically indistinguishable for every class (consistent with inherent class homogeneity, not train/test leakage)

ok
- **resplit_falsification**
random-split accuracy=1.0000, grouped-split accuracy=1.0000 (drop=+0.0000); grouped split reproduces random-split accuracy, consistent with inherent class homogeneity rather than a split-artifact explanation

ok
- **scenario_holdout_falsification**
no schema.scenario_column configured

ok
- **class_distribution_report**
5 classes, train imbalance ratio (majority:minority) = 24465:1; classes below 1% of train: ['Normal', 'Theft']

warning
- **low_cardinality_warning**
unique values by column: {'saddr': 21, 'daddr': 81, 'sport': 65541, 'dport': 6914}. ['saddr'] have fewer than 50 unique values, risk of memorizing specific hosts rather than learning attack behavior

warning
- **schema_fingerprint_check**
no known extractor signature matched

ok
- **data_integrity_check**
no missing labels, constant features, or +-inf values found

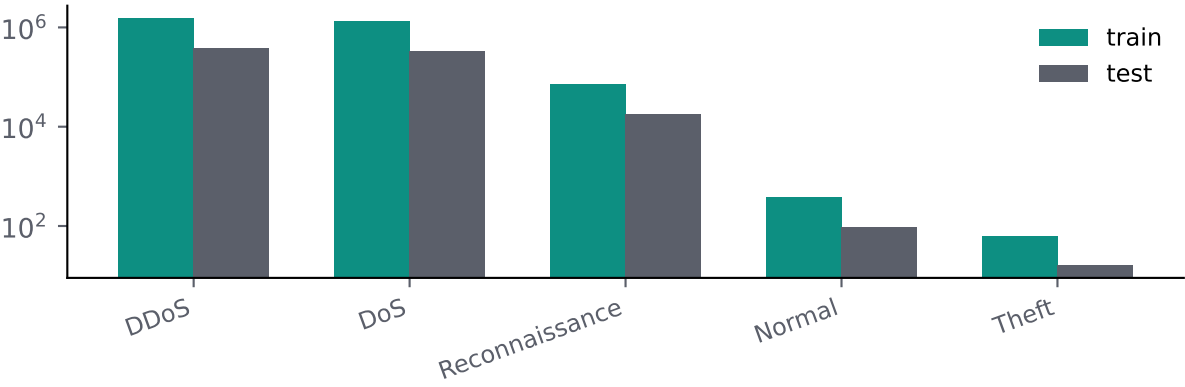
ok
- **result_robustness_check**
accuracy ranges from 1.0000 (grouped_split) to 1.0000 (random_split) across 3 conditions, a spread of 0.0000

ok
- **known_issue_lookup**
The full dataset is over 99.9% attack traffic (benign flows are 129,437 of 30,420,086 rows, 0.43%), an order of magnitude more skewed than most NIDS datasets; a uniform row-level sample may carry very few or zero benign rows, and a classifier's accuracy on this data mostly reflects its attack-vs-attack discrimination, not its ability to recognize normal traffic. (Koroniotis et al. 2019, Future Generation Computer Systems)

warning
- **repeated_seed_falsification_check**
grouped-split accuracy drop across 10 seeds: mean=+0.0000, 95% CI=[-0.0000, +0.0000]; not material: even the most generous estimate in the 95% CI (+0.0000) stays below the +0.0100 materiality threshold

ok

Class distribution (log scale)



Top feature importances (leakage_screen)

